

Jadavpur University



Computer Networks Lab Report

NAME : ANKIT SHAW

ROLL NO. : 002310501032

YEAR : BCSE UG – III (A1)

SESSION : 2023 – 27

ASSIGNMENT : 0 5

PROBLEM STATEMENT: Install Wireshark in local machine and do traffic and packet analysis according to the given questions

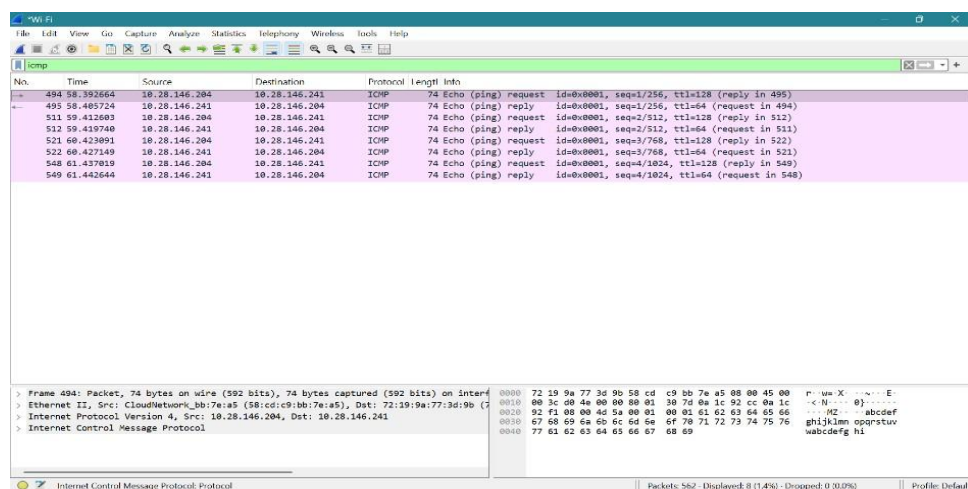
QUESTIONS:

- 1.) Generate some ICMP traffic by using the Ping command line tool to check the connectivity of a neighbouring machine (or router). Note the results in Wireshark. The initial ARP request broadcast from your PC determines the physical MAC address of the network IP Address, and the ARP reply from the neighbouring system. After the ARP request, the pings (ICMP echo request and replies) can be seen.

```
Pinging 10.28.146.241 with 32 bytes of data:
Reply from 10.28.146.241: bytes=32 time=13ms TTL=64
Reply from 10.28.146.241: bytes=32 time=7ms TTL=64
Reply from 10.28.146.241: bytes=32 time=4ms TTL=64
Reply from 10.28.146.241: bytes=32 time=5ms TTL=64

Ping statistics for 10.28.146.241:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 13ms, Average = 7ms
```

PING



The screenshot shows a Wireshark packet capture on the 'Wi-Fi' interface. The packet list on the left shows four ICMP Echo (ping) requests and four replies. The selected packet (No. 494) is an ICMP Echo (ping) request from 10.28.146.204 to 10.28.146.241. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
494	58.592664	10.28.146.204	10.28.146.241	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 495)
495	58.485724	10.28.146.241	10.28.146.204	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 494)
511	59.412693	10.28.146.204	10.28.146.241	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 512)
512	59.419740	10.28.146.241	10.28.146.204	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 511)
521	60.423891	10.28.146.204	10.28.146.241	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 521)
522	60.427149	10.28.146.241	10.28.146.204	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 521)
548	61.437819	10.28.146.204	10.28.146.241	ICMP	74	Echo (ping) request id=0x0001, seq=4/1824, ttl=128 (reply in 549)
549	61.442644	10.28.146.241	10.28.146.204	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1824, ttl=64 (request in 548)

Frame 494: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
> Ethernet II, Src: CloudNetwork_b7e:a5 (58:cd:c9:bb:7e:a5), Dst: 72:19:9a:77:3d:9b (7:8d:66:4c:8b:40)
> Internet Protocol Version 4, Src: 10.28.146.204, Dst: 10.28.146.241
> Internet Control Message Protocol

0000 72 19 9a 77 3d 9b 58 cd c9 bb 7e a5 08 00 45 00 n:waX...E
0010 00 3c 00 4a 00 00 59 01 30 7d 0a 1c 92 cc 0a 1c <N...0)....
0020 92 f1 08 00 4d 5a 00 01 00 01 61 62 63 64 65 66 ...M2...abdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmnopqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabdefgh

ICMP Requests

No.	Time	Source	Destination	Protocol	Length	Info
7	0.252519	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
73	5.782218	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
76	6.886114	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
83	7.831265	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
189	13.155580	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
118	14.378881	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
129	15.203849	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
175	20.732683	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
193	22.588249	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
192	22.588288	CloudNetwork_b6:7e...	72:19:9a:77:3d:9b	ARP	42	10.28.146.204 is at 58:c6:c9:80:7a:a5
193	22.781646	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
208	28.105898	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
224	29.123206	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
318	35.477555	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
315	36.501779	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
318	37.525863	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
348	43.260395	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
367	44.284475	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
372	45.923323	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
413	51.149265	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
413	51.149348	CloudNetwork_b6:7e...	72:19:9a:77:3d:9b	ARP	42	10.28.146.204 is at 58:c6:c9:80:7a:a5
465	52.680663	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
484	58.005358	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
502	59.030791	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241
513	60.054852	72:19:9a:77:3d:9b	Broadcast	ARP	42	Who has 10.28.146.125? Tell 10.28.146.241

Frame 2: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface
 Ethernet II, Src: 72:19:9a:77:3d:9b (72:19:9a:77:3d:9b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Address Resolution Protocol (request)

ARP Reply

2.)a. find the list the different protocols that appear in the protocol column in the unfiltered packet-listing window of Wireshark.

No.	Time	Source	Destination	Protocol	Length	Info
177	4.454724	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	60	51168 -> 50 [SYN] Seq=6 Wm=65535 Len=0 MSS=1300 SACK_PERM=1
179	4.161179	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51164 [ACK] Seq=7725 Ack=1550 Wm=68032 Len=0
180	4.161179	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51164 [ACK] Seq=7725 Ack=2544 Wm=267264 Len=0
181	4.161179	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51164 [ACK] Seq=7725 Ack=2779 Wm=267264 Len=0
182	4.279311	10.28.146.204	10.28.146.241	DNS	80	Standard query 0x0836 A fonts.googleapis.com
183	4.279541	10.28.146.204	10.28.146.241	DNS	80	Standard query 0x0836 AAAA fonts.googleapis.com
184	4.291373	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TLSv1.3	434	Application Data
185	4.331645	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	51164 -> 443 [ACK] Seq=2739 Ack=8085 Wm=65824 Len=0
186	4.342540	10.28.146.241	10.28.146.204	DNS	96	Standard query response 0x0836 A fonts.googleapis.com A 142.250.67.74
187	4.342540	2001:1458:201:a4::1	2404:6800:4002:8001::1	TCP	80	80 -> 51168 [SYN, ACK] Seq=1 Ack=1 Wm=3120 Len=0 MSS=1300 SACK_PERM=1
188	4.342780	2404:6800:4002:8001::2404:6800:4002:8001::1	2001:1458:201:a4::1	TCP	74	51168 -> 80 [ACK] Seq=1 Ack=1 Wm=65200 Len=0
189	4.343152	2404:6800:4002:8001::2404:6800:4002:8001::1	2001:1458:201:a4::1	HTTP	418	GET /favicon.ico HTTP/1.1
190	4.348968	10.28.146.241	10.28.146.204	DNS	168	Standard query response 0x0836 AAAA fonts.googleapis.com AAAA 2404:6800:4002:8001::2004
192	4.350822	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	80	51172 -> 443 [SYN] Seq=6 Wm=65535 Len=0 MSS=1300 SACK_PERM=1
192	4.587155	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	80	443 -> 51172 [SYN, ACK] Seq=6 Ack=1 Wm=65535 Len=0 MSS=1300 SACK_PERM=1
193	4.587340	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	51172 -> 443 [ACK] Seq=1 Ack=1 Wm=65200 Len=0
194	4.586565	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	1374	51172 -> 443 [ACK] Seq=1 Ack=1 Wm=65200 Len=1300 [TCP PDU reassembled in 195]
195	4.586565	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	256	Client Hello (SHA1=fonts.googleapis.com)
196	4.610431	2001:1458:201:a4::1	2404:6800:4002:8001::1	TCP	74	80 -> 51168 [ACK] Seq=1 Ack=345 Wm=32896 Len=0
197	4.610431	2001:1458:201:a4::1	2404:6800:4002:8001::1	TCP	1374	80 -> 51168 [ACK] Seq=1 Ack=345 Wm=32896 Len=1300 [TCP PDU reassembled in 198]
198	4.610431	2001:1458:201:a4::1	2404:6800:4002:8001::1	HTTP	428	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
199	4.610431	2001:1458:201:a4::1	2404:6800:4002:8001::1	TCP	74	80 -> 51168 [FIN, ACK] Seq=1655 Ack=345 Wm=32896 Len=0
200	4.610431	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51172 [ACK] Seq=1 Ack=181 Wm=27284 Len=0
201	4.610431	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51172 [ACK] Seq=1 Ack=181 Wm=27284 Len=0
202	4.610431	2404:6800:4002:8001::2404:6800:4002:8001::1	2404:6800:4002:8001::1	TCP	74	443 -> 51172 [ACK] Seq=1 Ack=181 Wm=27284 Len=0

Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface
 Ethernet II, Src: CloudNetwork_b6:7e... (58:c6:c9:80:7a:a5), Dst: 72:19:9a:77:3d:9b (72:19:9a:77:3d:9b)
 Internet Protocol Version 6, Src: 2404:6800:4002:8001::2404:6800:4002:8001::1, Dst: 2404:6800:4002:8001::1
 Transmission Control Protocol, Src Port: 51168, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

List of Protocols

b. How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received? (By default, the value of the Time column in the packetlisting window is the amount of time, in seconds, since Wireshark tracing began. To display the Time field in time-of-day format, select the Wireshark View pull down menu, then select Time Display Format, then select Time-of-day.)

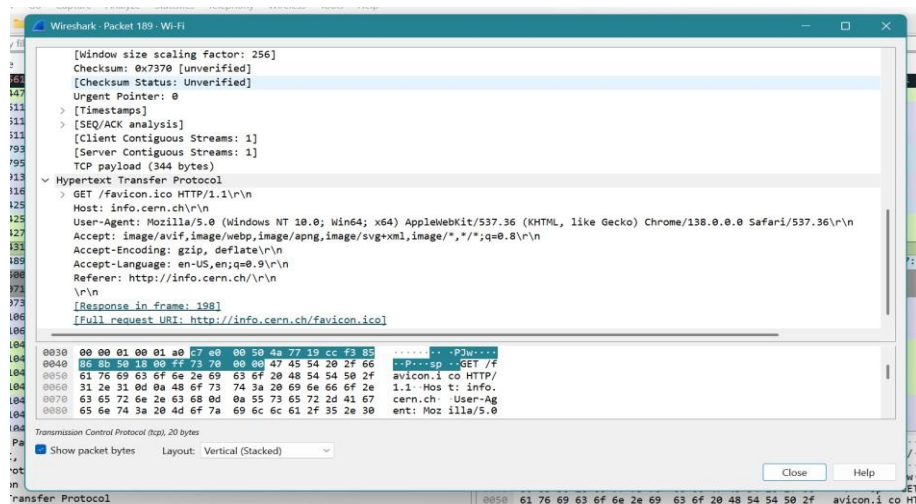
Ans. As we can see, that GET message was sent in time 4.343152 and OK response was received in time 4.610431. So the time taken is 0.267279 seconds.

c. What is the Internet address of the website? What is the Internet address of your computer?

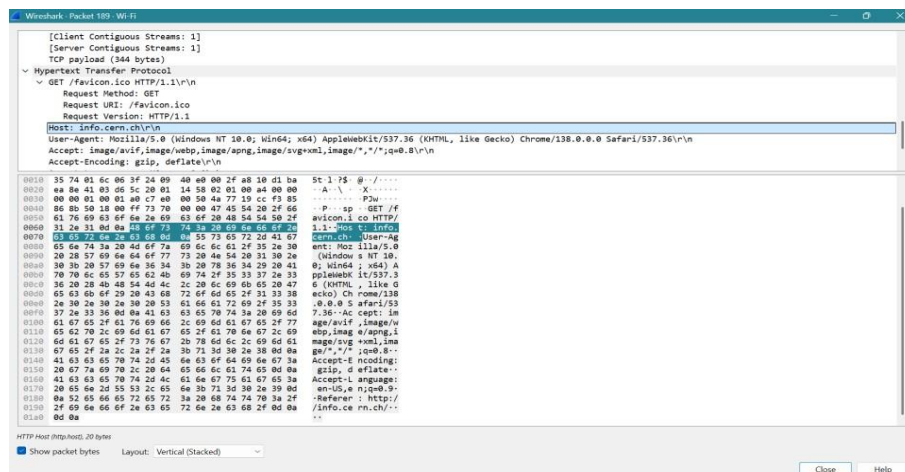
From the previous screenshot, we can see that the internet(IP) address of the website is – 136.232.79.144 and IP address of my computer is – 192.168.29.243.

d. Search back through your capture, and find an HTTP packet containing a GET command. Click on the packet in the Packet List Panel. Then expand the HTTP layer in the Packet Details Panel, from the packet.

e. Find out the value of the Host from the Packet Details Panel, within the GET command.



3.) Highlight the Hex and ASCII representations of the packet in the Packet Bytes Panel

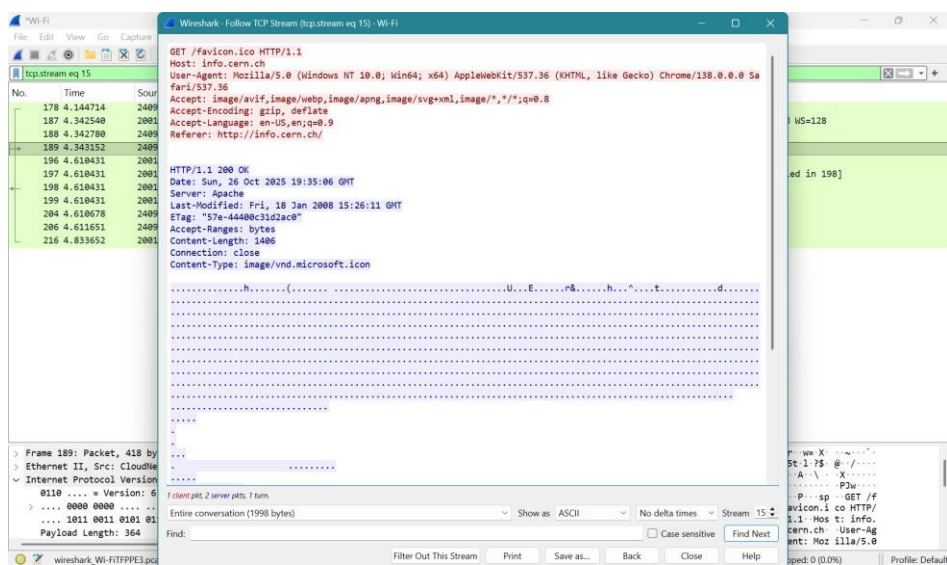


4.) Find out the first 4 bytes of the Hex value of the Host parameter from the Packet Bytes Panel

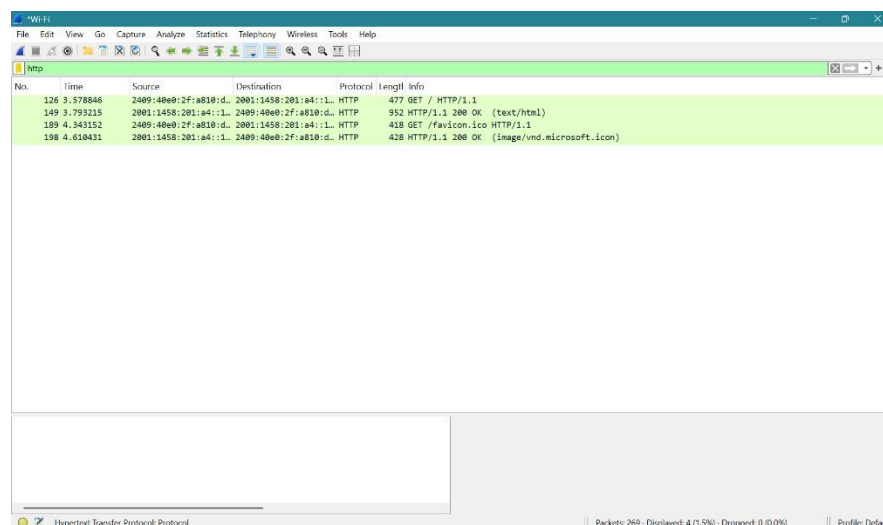
As we can see first four bytes of the Hex value of the Host parameter is: 48 6f 73 74

5. Filter packets with http, TCP, DNS and other protocols. a. Find out what are those packets contain by following one of the conversations (also called network flows), select one of the packets and press the right mouse button click on follow.

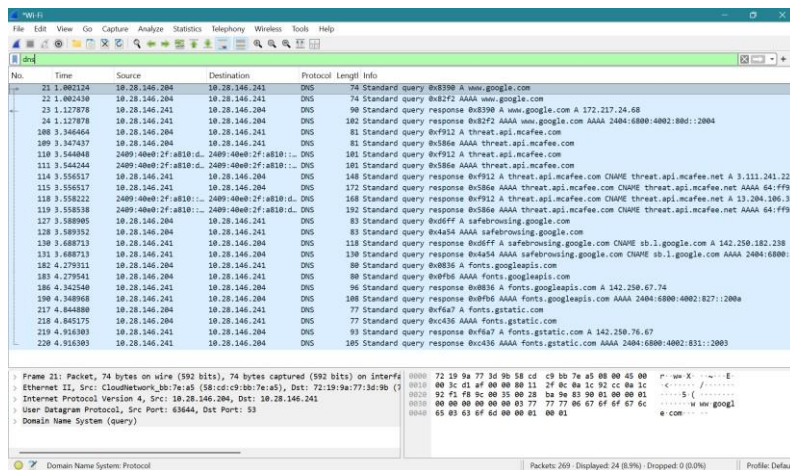
TCP Flow:



HTTP:

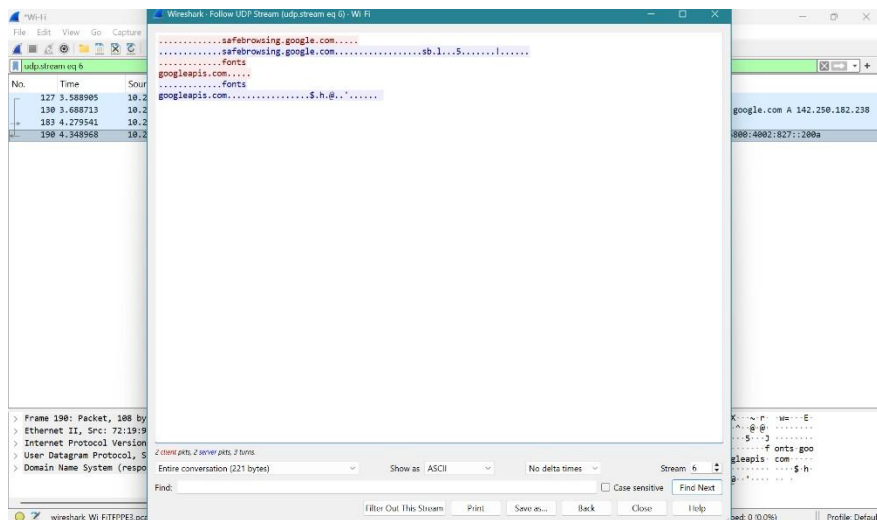


DNS:

A screenshot of a Wireshark capture showing a list of DNS packets. The top pane shows a list of packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The bottom pane shows the details of a selected packet, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (Query).

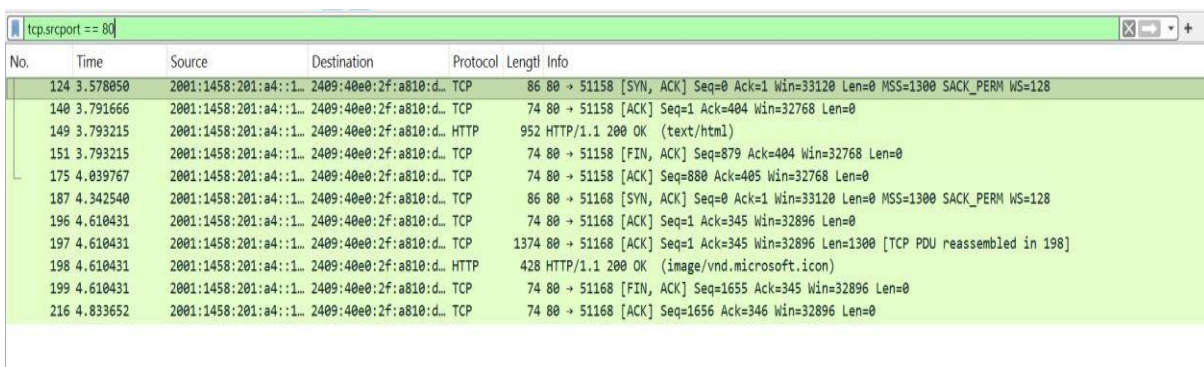
No.	Time	Source	Destination	Protocol	Length	Info
21	3.982334	10.28.146.204	10.28.146.241	DNS	74	Standard query 0x8398 A www.google.com
22	3.602430	10.28.146.204	10.28.146.241	DNS	74	Standard query 0x82f2 AAAA www.google.com
23	3.127878	10.28.146.241	10.28.146.204	DNS	98	Standard query response 0x8398 A www.google.com A 172.217.24.58
24	3.127878	10.28.146.241	10.28.146.204	DNS	162	Standard query response 0x82f2 AAAA www.google.com AAAA 2404:6800:4002:80d::2004
108	3.346464	10.28.146.204	10.28.146.241	DNS	81	Standard query 0xf912 A threat.api.mcafee.com
109	3.347437	10.28.146.204	10.28.146.241	DNS	81	Standard query 0x586e AAAA threat.api.mcafee.com
110	3.544048	2409:40e0:2f:a810:d...	2409:40e0:2f:a810:d...	DNS	161	Standard query 0xf912 A threat.api.mcafee.com
111	3.544244	2409:40e0:2f:a810:d...	2409:40e0:2f:a810:d...	DNS	161	Standard query 0x586e AAAA threat.api.mcafee.com
114	3.556517	10.28.146.241	10.28.146.204	DNS	148	Standard query response 0xf912 A threat.api.mcafee.com CNAME threat.api.mcafee.net A 3.111.241.22.
115	3.556517	10.28.146.241	10.28.146.204	DNS	172	Standard query response 0x586e AAAA threat.api.mcafee.com CNAME threat.api.mcafee.net AAAA 64:ff9...
118	3.558222	2409:40e0:2f:a810:d...	2409:40e0:2f:a810:d...	DNS	168	Standard query response 0xf912 A threat.api.mcafee.com CNAME threat.api.mcafee.net A 13.204.196.3.
119	3.558338	2409:40e0:2f:a810:d...	2409:40e0:2f:a810:d...	DNS	192	Standard query response 0x586e AAAA threat.api.mcafee.com CNAME threat.api.mcafee.net AAAA 64:ff9...
127	3.588905	10.28.146.204	10.28.146.241	DNS	83	Standard query 0xd0ff A safebrowsing.google.com
128	3.589312	10.28.146.204	10.28.146.241	DNS	83	Standard query 0xa454 AAAA safebrowsing.google.com
130	3.688713	10.28.146.241	10.28.146.204	DNS	118	Standard query response 0xd0ff A safebrowsing.google.com CNAME sb.1.google.com A 142.250.182.238
131	3.688713	10.28.146.241	10.28.146.204	DNS	130	Standard query response 0xa454 AAAA safebrowsing.google.com CNAME sb.1.google.com AAAA 2404:6800...
182	4.279511	10.28.146.204	10.28.146.241	DNS	80	Standard query 0xb036 A fonts.googleapis.com
183	4.279541	10.28.146.204	10.28.146.241	DNS	80	Standard query 0xb0f6 AAAA fonts.googleapis.com
186	4.342540	10.28.146.241	10.28.146.204	DNS	96	Standard query response 0xb036 A fonts.googleapis.com A 142.250.67.74
190	4.408958	10.28.146.241	10.28.146.204	DNS	168	Standard query response 0xb0f6 AAAA fonts.googleapis.com AAAA 2404:6800:4002:827::200a
217	4.848880	10.28.146.204	10.28.146.241	DNS	77	Standard query 0xbf6a7 A fonts.gstatic.com
218	4.845175	10.28.146.204	10.28.146.241	DNS	77	Standard query 0xc436 AAAA fonts.gstatic.com
219	4.916303	10.28.146.241	10.28.146.204	DNS	93	Standard query response 0xbf6a7 A fonts.gstatic.com A 142.250.76.67
220	4.916303	10.28.146.241	10.28.146.204	DNS	185	Standard query response 0xc436 AAAA fonts.gstatic.com AAAA 2404:6800:4002:831::2003

UDP Flow:

A screenshot of Wireshark showing a UDP stream analysis. The left pane shows a list of packets. The middle pane shows the details of a selected packet, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (Response). The right pane shows the raw data of the packet.

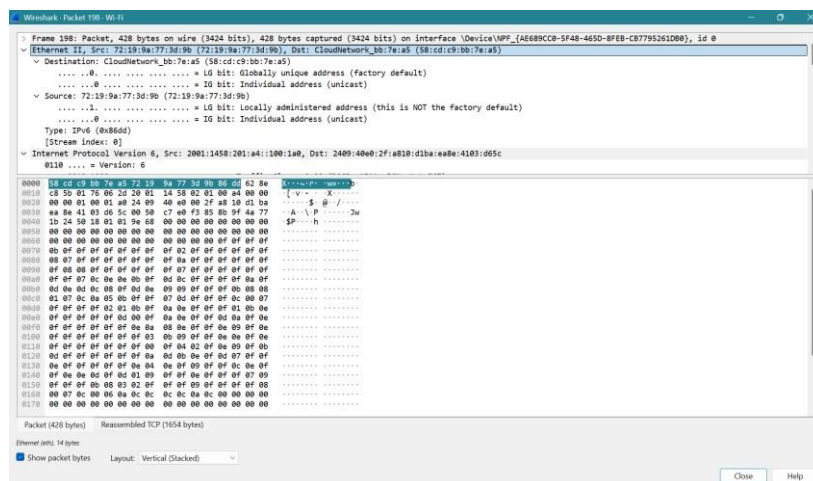
No.	Time	Source	Destination	Protocol	Length	Info
127	3.588905	10.2	10.2	DNS	83	Standard query 0xd0ff A safebrowsing.google.com
130	3.688713	10.2	10.2	DNS	118	Standard query response 0xd0ff A safebrowsing.google.com CNAME sb.1.google.com A 142.250.182.238
183	4.279541	10.2	10.2	DNS	80	Standard query 0xb036 A fonts.googleapis.com
190	4.408958	10.2	10.2	DNS	168	Standard query response 0xb0f6 AAAA fonts.googleapis.com AAAA 2404:6800:4002:827::200a

6. Search through your capture, and find an HTTP packet coming back from the server (TCP Source Port == 80). Expand the Ethernet layer in the Packet Details Panel. TCP Source Port == 80

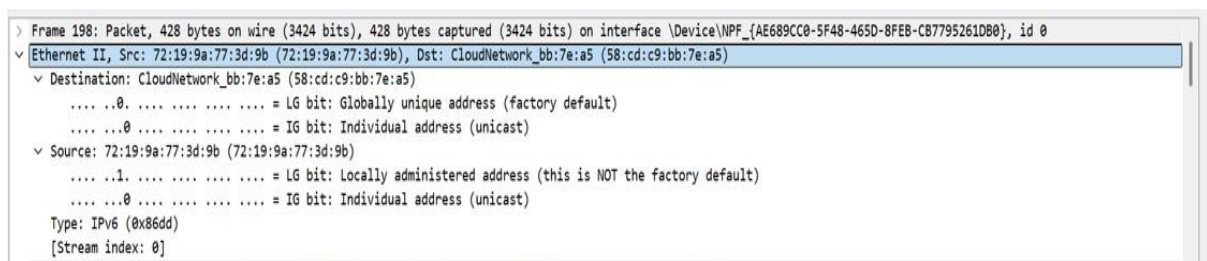
A screenshot of a Wireshark capture showing a list of packets. The top pane shows a list of packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The bottom pane shows the details of a selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
124	3.578850	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	86	80 → 51158 [SYN, ACK] Seq=0 Ack=1 Win=33120 Len=0 MSS=1300 SACK_PERM WS=128
140	3.791666	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51158 [ACK] Seq=1 Ack=404 Win=32768 Len=0
143	3.793215	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	HTTP	952	HTTP/1.1 200 OK (text/html)
151	3.793215	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51158 [FIN, ACK] Seq=879 Ack=404 Win=32768 Len=0
175	4.039767	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51158 [ACK] Seq=880 Ack=405 Win=32768 Len=0
187	4.342540	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	86	80 → 51168 [SYN, ACK] Seq=0 Ack=1 Win=33120 Len=0 MSS=1300 SACK_PERM WS=128
196	4.610431	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51168 [ACK] Seq=1 Ack=345 Win=32896 Len=0
197	4.610431	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	1374	80 → 51168 [ACK] Seq=1 Ack=345 Win=32896 Len=1300 [TCP PDU reassembled in 198]
198	4.610431	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	HTTP	428	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
199	4.610431	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51168 [FIN, ACK] Seq=1655 Ack=345 Win=32896 Len=0
216	4.833652	2001:1458:201:a4::1	2409:40e0:2f:a810:d...	TCP	74	80 → 51168 [ACK] Seq=1656 Ack=346 Win=32896 Len=0

Ethernet Layer Details:



7.) What are the manufacturers of your PC's Network Interface Card (NIC), and the servers NIC?



Source – 72:19:9a:77:3d:9b 72:19:91

Destination (server) 58:cd:c9:bb:7e:a5 58:cd:c9 Cloudflare, Inc.

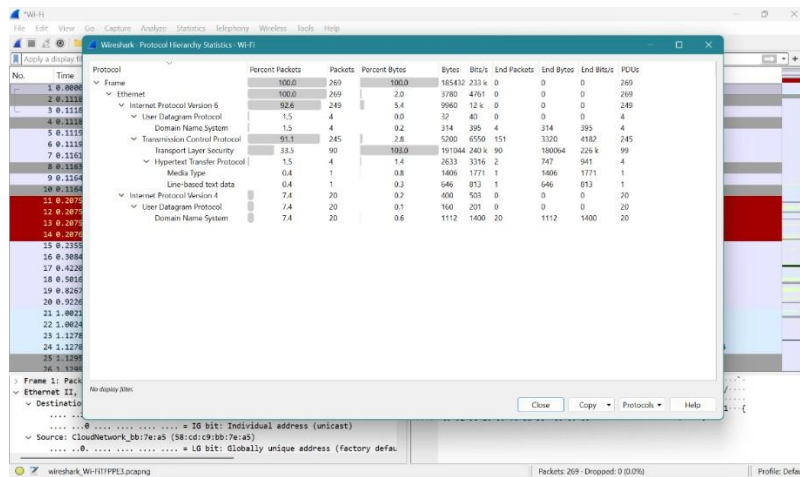
8.) What are the Hex values (shown in the raw bytes panel) of the two NICs Manufacturers OUIs?

For Laptop's Manufacturer :- 72:19:9a:77:3d:9b → Locally Administered (no real manufacturer)

For Server's Manufacturer :- 58:cd:c9:bb:7e:a5 → Cloudflare, Inc

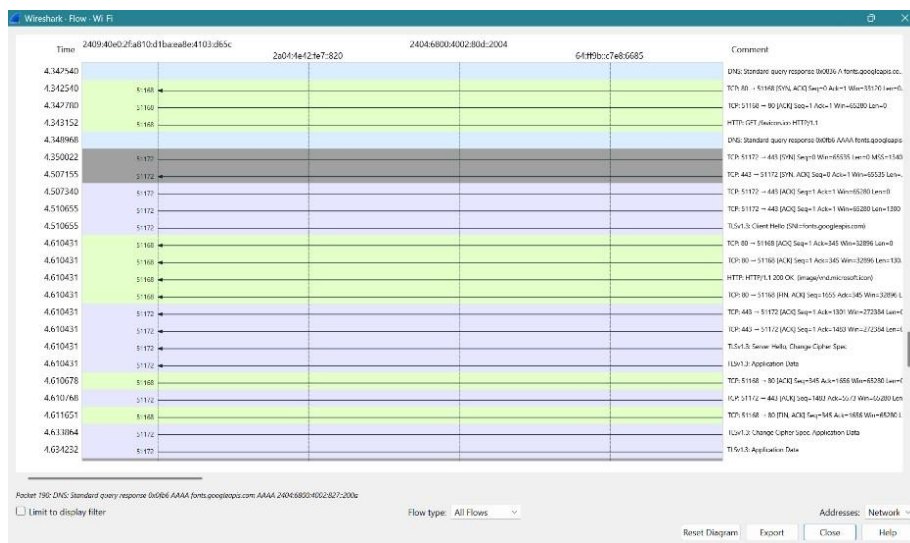
9. Find the following statistics: a. What percentage of packets in your capture are TCP, and give an example of the higher level protocol which uses TCP?

b. What percentage of packets in your capture are UDP, and give an example of the higher level protocol which uses UDP?



Network Stats

9.) Find the traffic flow Select the Statistics->Flow Graph menu option. Choose General Flow and Network Source options, and click the OK button.



Traffic Flow